Certified Kubernetes Administrator (CKA): Managing Cluster Lifecycle and Upgrades

Backing up and restoring etcd for disaster recovery

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Four framing questions

- What does etcd database actually store, and why is it the single source of truth?
- How do I take an etcd snapshot, and which binary does the work?
- What's the full restore workflow when disaster strikes?
- How do stacked and external etcd topologies compare?

Checking in with Globomantics

"It's 2 a.m., and the on-call engineer just deleted the production namespace. Deployments, Services, ConfigMaps -- all gone. The CTO is on Slack with one question: do we have a backup?"

Kai S., On-Call Site Recovery Engineer (SRE), Globomantics Robotics Company

What does etcd store, and why is it the single source of truth?

etcd: The cluster's single source of truth

Every Kubernetes object is serialized and stored in etcd

Deployments, Services, ConfigMaps, Secrets, RBAC rules -- all of it

etcd is a distributed key-value store that lives in the control plane

etcd: The cluster's single source of truth

Lose etcd without a backup and the cluster's declared state is gone

etcdctl speaks the v3 API, the default since etcd 3.4

Only the API server reads and writes etcd directly

How do I take an etcd snapshot, and which binary does the work?

Taking an etcd snapshot with etcdctl

--endpoints=https://127.0.0.1:2379

- targets the local etcd

--cacert

- verifies the etcd server's identity

etcdctl snapshot save

- takes a live, point-in-time backup

save

- stays in etcdctl because it talks to a running cluster

Verify the file

- with etcdutl snapshot status, not etcdctl

--cert and --key

- are your client credentials for mutual TLS

```
# Take a live snapshot -- ONLINE op, stays in etcdctl, needs mutual TLS
ETCDCTL_API=3 etcdctl snapshot save /tmp/etcd-backup.db \
  --endpoints=https://127.0.0.1:2379 \
  --cacert=/etc/kubernetes/pki/etcd/ca.crt \
  --cert=/etc/kubernetes/pki/etcd/server.crt \
  --key=/etc/kubernetes/pki/etcd/server.key

# Verify the file -- OFFLINE op, moved to etcdutl in etcd 3.6
etcdutl snapshot status /tmp/etcd-backup.db --write-out=table
```

Snapshot save with etcdctl, verify with etcdutl

What's the full restore workflow when disaster strikes?

The etcd restore workflow, step by step

Stop the API server: move kube-apiserver.yaml out of /etc/kubernetes/manifests/

Restore to a NEW data directory with etcdutl snapshot restore

Point etcd.yaml's hostPath at the restored directory

Move kube-apiserver.yaml back so kubelet restarts it

Verify with kubectl get nodes, then confirm the deleted objects are back

Restore from a snapshot with etcdutl

```
# 1) Stop the API server (static Pod) by moving its manifest aside
sudo mv /etc/kubernetes/manifests/kube-apiserver.yaml /tmp/

# 2) Restore into a NEW data dir -- OFFLINE op, etcdutl (no TLS flags)
sudo etcdutl snapshot restore /tmp/etcd-backup.db \
  --data-dir=/var/lib/etcd-restored

# 3) Repoint etcd at the restored data dir (the step folks forget)
sudo sed -i 's#/var/lib/etcd$#/var/lib/etcd-restored#' \
  /etc/kubernetes/manifests/etcd.yaml

# 4) Bring the API server back; kubelet restarts etcd + apiserver
sudo mv /tmp/kube-apiserver.yaml /etc/kubernetes/manifests/

# 5) Confirm the cluster and the recovered objects
kubectl get nodes
kubectl get all -n globomantics-shop
```

How do stacked and external etcd topologies compare?

Stacked vs. external etcd topologies

Stacked

Stacked is kubeadm's default -- fewer nodes, simpler to stand up

Stacked couples failure: lose the node, lose a control-plane AND etcd member

Stacked: etcd runs on each control-plane node, next to the API server

vs.

External

External isolates the etcd failure domain and is more resilient

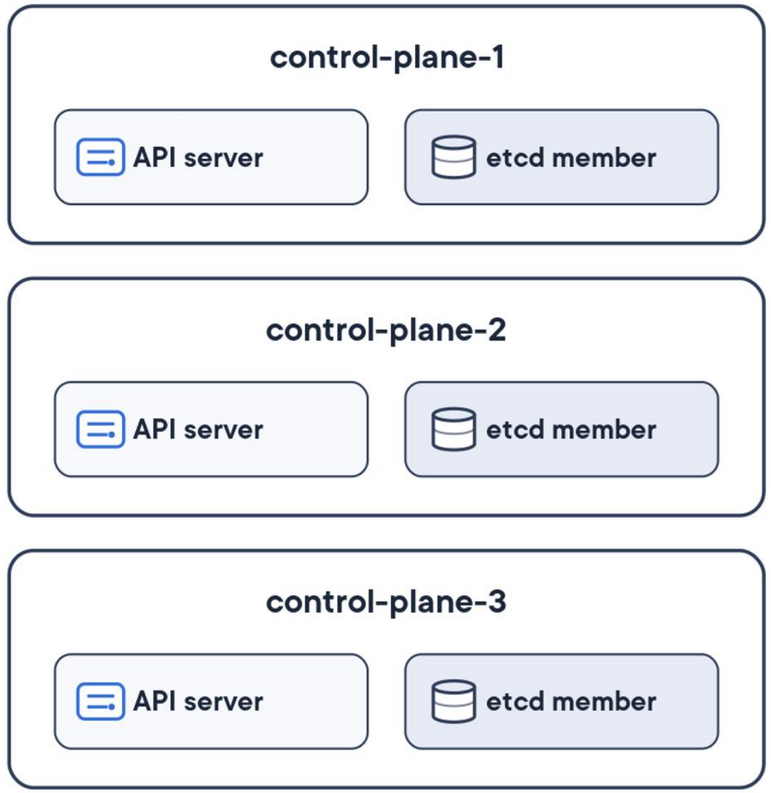
External costs more nodes to provision, patch, monitor, and back up

External: etcd runs on its own dedicated nodes, off the control plane

Stacked vs. external etcd: one picture

STACKED

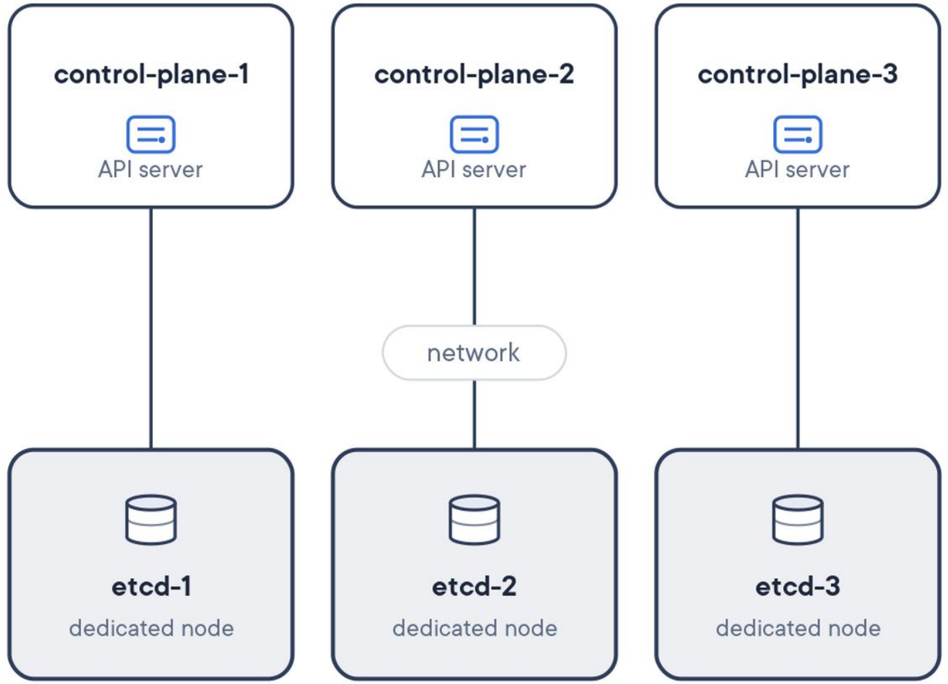
kubeadm default



Simpler, fewer nodes -- but a lost node takes a control-plane AND an etcd member with it.

EXTERNAL

dedicated etcd nodes



Resilient -- isolates the etcd failure domain -- but you provision, patch, and back up more nodes.

Demo

Back up, destroy, and restore etcd

Snapshot the live Vagrant VM cluster, delete a namespace, and bring it back with etcdctl and etcdutl.

Checking out with Globomantics

"The restore took four minutes. Four minutes from total disaster to a healthy cluster. The CTO stopped asking whether we had a backup and started asking how we put one on every cluster."

Kai S., On-Call Site Recovery Engineer (SRE), Globomantics Robotics Company

From Globomantics to you

etcd is the single source of truth

protect it or you can't recover

snapshot save lives in etcdctl

status and restore moved to etcdutl in 3.6

Drill the restore

stop API server, restore to a new dir, repoint etcd, verify

Know both topologies

stacked is simple, external isolates failure